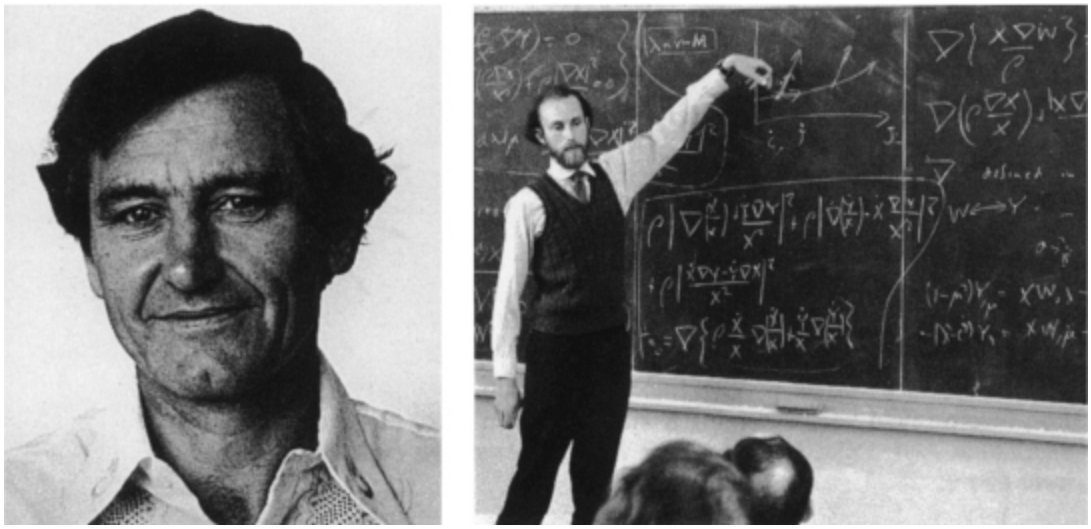
The effects of spin on a black hole were not understood until the late 1960s. The understanding came largely from Brandon Carter, one of Dennis Sciama's students at Cambridge University.

When Carter joined Sciama's group in autumn 1964, Sciama immediately suggested, as his first research problem, a study of the implosion of realistic, spinning stars. Sciama explained that all previous calculations of implosion had dealt with idealized, nonspinning stars, but that the time and tools now seemed right for an assault on the effects of spin. A New Zealander mathematician named Roy Kerr had just published a paper giving a solution of Einstein's field equation that seemed to describe the spacetime curvature outside a spinning star. This was the first solution for a spinning star that anyone had ever found. Unfortunately, Sciama explained, it was a very special solution; it surely could not describe *all* spinning stars. Spinning stars have lots of "hair" (lots of properties such as complicated shapes and complicated internal motions of their gas), and Kerr's solution did not have much "hair" at all: The shapes of its spacetime curvature were very smooth, very simple; too simple to correspond to typical spinning stars. Nevertheless, Kerr's solution of Einstein's field equation was a place to start.



Left: **Roy Kerr ca. 1975.** ***Right:*** **Brandon Carter lecturing about black holes at a summer school in the French Alps in June 1972.** [Left: courtesy Roy Kerr; right: photo by Kip Thorne.]

Few research problems have the immediate payoff that this one did: Within a year Carter had shown mathematically that Kerr's solution describes not a spinning star, but rather a spinning black hole. (This